



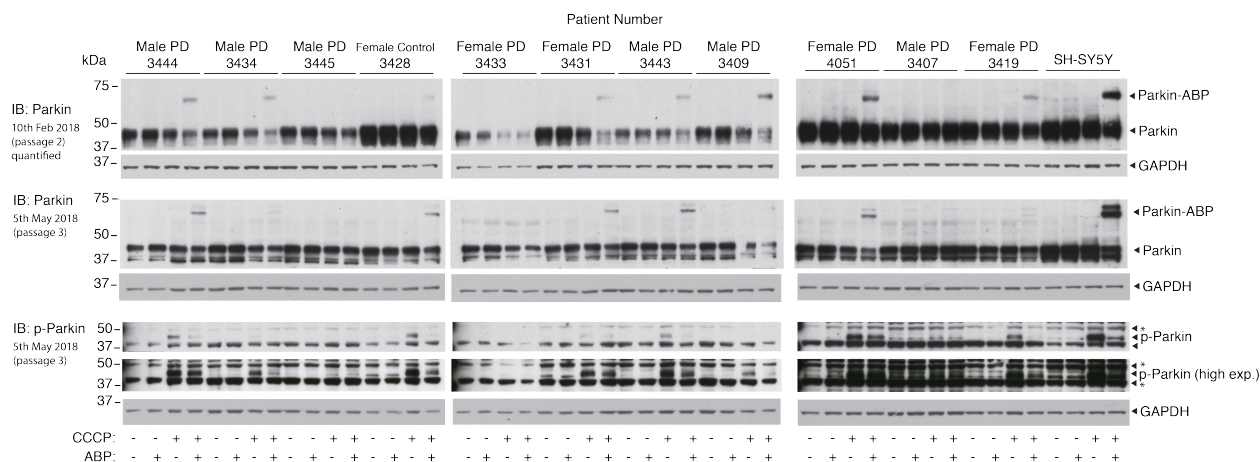
Chemical Probe Profiling of Parkin Activity in PPMI Fibroblast Samples

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Eleven blinded fibroblast samples from the PPMI were successfully cultured in the presence and absence of CCCP for 3 hours. Lysates were obtained and were treated with a Parkin activity-based probe that provides a measure of the enzymes ability to covalent accept ubiquitin from the upstream E2 enzyme¹. Probe labelling was assessed by Parkin bandshift and samples were immunoblotted with anti-parkin (Santa Cruz, 1:500) and anti-phosphoparkin (Muqit, 1:2500) antibodies.

Eight of the 11 samples demonstrated Parkin activity as inferred from the Parkin activity-based probe assay. In all cases, Parkin phosphorylation correlated with probe labelling suggesting that Parkin phosphorylation might serve as a robust biomarker for Parkin activation. Interestingly, the stability of Parkin after CCCP treatment varied substantially across different active samples. This raises the possibility that Parkin activation in response to mitochondrial insult is not sufficient and pathogenesis can arise from Parkin that is functional but has reduced lifetime upon activation. Furthermore, sample 11400841 was strikingly destabilized upon CCCP treatment but did not display any detectable probe labelling signals nor Parkin phosphorylation. This cell line also demonstrated retarded growth relative to other samples so it is not clear if this is related.





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NB. Double Parkin-ABP band observed in 5th May 2018 experiment is because 5 uM EDTA is present in the lysis buffer. We find that this serves as a Parkin activator and was added in the 5th May experiment. We find that SH-SY5Y control cell labelling is enhanced, but still dependent on CCCP, in the presence of EDTA but double labelling is observed in this instance. This is most likely due to Zn ion extraction imparting a structural defect.





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PATNO	Parkin activity (approximate % probe labelling) Passage 2 cells 10 th Feb18	Parkin phosphorylation Passage 2 cells 10 th Feb18	Parkin activity (approximate % probe labelling) Passage 3 cells 5 ^h May18	Parkin phosphorylation Passage 3 cells 5 ^h May18	Notes
3444	19	Yes	20	Yes	
3434	21	Yes	6	Yes	
3445	inactive	No	inactive	No	
3428	5	Yes	16	Yes	
3433	Inactive	No	Inactive	No	Slow cell growth Striking Parkin destabilization upon CCCP treatment
3431	15	Yes	15	Yes	
3443	20	Yes	25	Yes	
3409	37	Yes	Not detectable due to air bubble	Yes	
4051	5	Yes	29	Yes	
3407	Inactive	No	Inactive	No	
3419	3	Yes	6	Yes	
SH-SY5Y	27	Yes	70	Yes	Improved labelling efficiency in passage 3 cells attributed to presence of the EDTA activator in the lysis buffer





Method

PPMI fibroblast vials were thawed and contents dispensed into fresh medium in 10 cm dishes. Culture medium (10 ml) was composed of DMEM (Thermofisher) supplemented with 15 % FBS, 2.0 mM L-glutamine, 1× minimum essential medium (MEM, Life Technologies), 1× non-essential amino acids (NEAA, Life Technologies), 1.0 mM of sodium pyruvate (Life Technologies) and antibiotics (100 units mL⁻¹ penicillin, 0.1 mg mL⁻¹ streptomycin. Cells were grown in an incubator (37 °C, 5% CO₂) until they reached 80% confluency (~2 days). NB. Sample #11400841 grew significantly slower than other samples (~3 days to reach 80% confluency). Cells were detached with trypsin and seeded into 5 x 10 cm dishes. Cells were grown to 80% confluency (~3 days) yielding passage 2 cells (cells from 2 of these dishes were detached and stored at -80 C and one dish was seeded into 2 x 10 cm dishes and cultured yielding passage 3 cells). One dish was treated with CCCP (10 uM) and one was treated with DMSO (labelled as experiment 10th Feb 18). Cells were incubated for a further 3h and then lysed using extensively degassed lysis buffer (50 mM Na₂HPO₄-HCl pH 7.5, 0.27 M sucrose, 10 mM sodium 2-glycerophosphate, 0.2 mM phenylmethane sulfonyl fluoride (PMSF), 1.0 mM benzimidazole, 1.0 mM sodium orthovanadate, 50 mM sodium fluoride, 5.0 mM sodium pyrophosphate and 10 µM TCEP). Lysate concentrations were measured by Bradford assay and normalized to 1.3 mg/ml. Lysate samples (40 uL) were mixed with 115 uM stock of ABP 8¹ to a final concentration of 5 uM . Reaction was incubated for 4h at 30 C. Samples were then mixed with reducing 4X LDS sample buffer. Unboiled samples were resolved by 4-12% Bis-Tris Nupage SDS-PAGE with MOPS running buffer (25-30 ug loading per well). Protein was then transferred to nitrocellulose membrane and probed with total and phosphoparkin antibodies as described previously¹. Phosphoparkin antibody required depletion against 2 x milk blocked nitrocellulose membranes and 1 x mock lysate (PARK2 deltaex5/C441R fibroblast sample¹) transferred membrane (overnight incubations at 4 C). Presented total Parkin data are from passage 2 cells. Phospho-parkin data was unsatisfactory for these cells due to inadequate antibody depletion so the experiment was repeated with passage 3 cells and the anti-Parkin and anti-Phospho-Parkin data from these experiments is presented (labelled as 5th May 18). In this experiment, 5 µM EDTA was added to the lysis buffer which increases Parkin activation in response to CCCP treatment. However, fibroblasts did not tolerate high passage as growth halted during passage 3. Probe labelling efficiency of Parkin also appeared to drop in passage 3 cells relative to passage 2 cells. For example, probe labelling signals are lower for some samples compared to the passage 2 experiments even though the EDTA activator is present. Probe labelling and Parkin phosphorylation trends were largely consistent across passage 2 and passage 3 experiments. Blots were visualized with ECL Prime reagent (GE Life Sciences). Fraction of probe labelling was approximately quantified by densitometry (ImageJ). Area of labelled band / area of labelled band + unlabeled band * 100.





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References

1. Pao, K. C. et al. Probes of ubiquitin E3 ligases enable systematic dissection of parkin activation. Nat. Chem. Biol. 12, 324–331 (2016).

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